SITTTR Revision 2021 · Semester 5 · Program Core

Electricity Generation, Transmission & Distribution

Complete handbook lesson for Course 5032. It follows the official syllabus: generating systems, economics, power factor, tariffs, transmission schemes, sag, distribution systems, insulators, underground cables and substations.

4 Credits
Program Core

60 Periods
L:3 T:1 P:0

4 Chapters
CO1 to CO4

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Course Information

Program: Diploma in Electrical and Electronics Engineering. Course code: 5032. Semester: 5. Credits: 4. Periods: 4 per week and 60 per semester.

Objectives

- Introduce electricity generation plants and their economics.
- Familiarize students with electrical transmission and distribution systems.

Course Outcomes

- CO1: Identify generating systems and calculate electricity generated.
- CO2: Choose power factor improvement methods and list tariffs.
- CO3: Identify transmission schemes and calculate sag.
- CO4: Summarize distribution systems.

Malayalam: □ □□□□□ generation □□□□ consumer distribution □□□ power system flow □□□□□□□□□□□□□□□□□.

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No narrow centered layout: chapters, formulas, diagrams and questions use the full page width.

Module 1 · CO1 · 14 Hours

Electric Power Generating Systems

Energy source classification, commercial/non-commercial generation, hydroelectric station, thermal stations, site selection, hydel calculations, interconnected grid, distributed generation and microgrid.

Energy sources and methods

Electricity is produced by converting water energy, fuel heat, nuclear heat or renewable energy into electrical output through a generator. Conventional sources include coal, diesel, gas, nuclear and hydro. Renewable sources include solar, wind, biomass and small hydro.

Malayalam: Water, coal, gas, diesel □□□□□□□□ sources mechanical rotation □□□□□ alternator □□□ electricity □□□□□□□□□□□□□.

Hydroelectric plant

Stored water flows through a penstock, rotates a turbine and drives an alternator. Hydel plants have low running cost and quick starting, but depend on rainfall and suitable site.

$$P = 9.81\,Q\,H\,\eta \text{ kW}$$

Thermal plants

Coal, gas, diesel and nuclear stations use heat energy to produce mechanical rotation. Coal and nuclear plants are base-load suited; gas and diesel are useful for peak or standby operation.

Grid and microgrid

Interconnected grids improve reliability and economy. Distributed generation places sources near loads. A microgrid can operate with the main grid or independently during grid failure.

Module 2 · CO2 · 15 Hours

Load Curve, Cost, Power Factor and Tariffs

Variable load, load curve, load duration curve, demand terms, generation costs, power factor correction and tariff schemes.

Load curve

Load curve shows demand versus time. The area under the curve gives energy. It helps decide base load, peak load, reserve capacity and economic operation.

Malayalam: Load curve demand □□□□□□ □□□□□□□□ □□□□□□□□□ □□□□□ □□□□□□□□□□□□□.

Demand terms

Connected load, average demand, maximum demand, load factor and diversity factor are used for power station and distribution planning.

> Load factor = Average demand / Maximum demand

Power factor improvement

Low power factor increases current, losses, voltage drop and equipment rating. Static capacitors, synchronous condensers and phase advancers improve power factor.

> $Q_c = P(\tan\varphi_1 - \tan\varphi_2)$

Tariffs

Simple, flat rate, block rate, two-part, three-part, maximum demand, power factor, ABT and TOD tariffs are used for billing and demand management.

Module 3 · CO3 · 15 Hours

Transmission Systems and Sag

Primary/secondary transmission, AC/DC transmission, overhead lines, conductors, losses, regulation, sag, transposition, skin effect, proximity effect, Ferranti effect, corona, EHV, HVDC and FACTS.

Transmission schemes

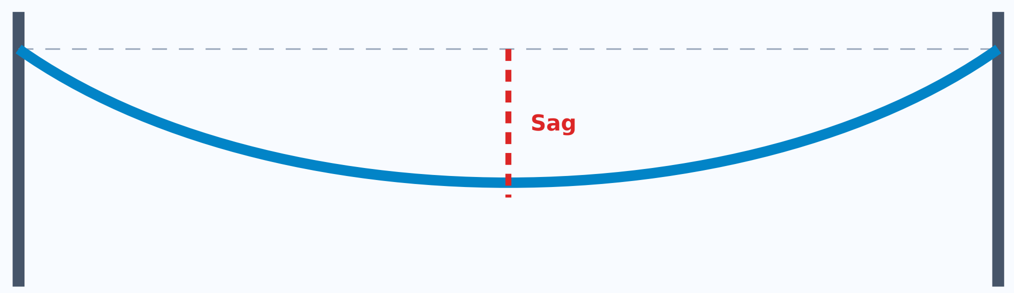
Transmission carries bulk power from generating stations to receiving stations. AC transmission is common due to easy transformation. DC transmission is useful for special long-distance and interconnection cases.

Overhead line components

Conductors, supports, insulators, cross-arms, clamps, earth wires and protective accessories form the overhead line system.

Sag and tension

Sag is the vertical distance between the lowest conductor point and the line joining supports. Correct sag gives safe clearance without excessive tension.



Sag equation for equal-level supports: $y = wL^2 / 8T$.

Module 4 · CO4 · 14 Hours

Distribution Systems, Insulators, Cables and Substations

Feeder, distributor, service mains, radial/ring/interconnected systems, DC/AC distribution, ideal system, insulators, string efficiency, underground cables and substations.

Distribution components

Feeder carries power from substation to the area. Distributor gives tappings to consumers. Service mains connect distributor to consumer premises.

Malayalam: Feeder supply area □□□, distributor consumers-□□□□□□ tapping, service mains final connection.

Distribution schemes

Radial system is simple but less reliable. Ring main improves continuity. Interconnected system provides high reliability but protection is complex.

Insulators and cables

Insulators support conductors and prevent leakage. Underground cables have conductor, insulation, sheath, bedding, armouring and serving.

> String efficiency = Vstring / (n × Vmax disc) × 100%

Substation

A substation changes voltage level, switches circuits, protects equipment and controls power flow using transformers, busbars, circuit breakers, isolators, CT, PT, relays and lightning arresters.

Formula Bank

P = 9.81QHη

Hydel power in kW.

Energy = Power × Time

kWh calculation.

LF = Average demand / Maximum demand

Load factor.

DF = Σ individual max demand / system max demand

Diversity factor.

$Q_c = P(\tan\varphi_1 - \tan\varphi_2)$

Capacitor kVAr.

$y = wL^2 / 8T$

Sag for equal supports.

Worked Examples

Hydel power

Q=20, H=40, η=0.85. P=9.81×20×40×0.85=6670.8 kW.

6.67 MW

Load factor

Average demand=300 kW, max demand=500 kW. LF=300/500.

60%

Capacitor rating

100 kW at 0.7 improved to 0.95. Qc=100(1.020−0.329).

69.1 kVAr

Sag

w=0.8 kg/m, L=200 m, T=2000 kg. $y=0.8\times200^2/(8\times2000)$.

2 m

Question Bank

2 and 5 Mark

1. Define load factor.

2. What is sag?
3. List two PF improvement methods.
4. Define feeder and distributor.
5. Explain load curve and its importance.
6. Compare radial and ring main distribution.
7. Explain skin effect and corona.
8. Write short note on underground cable construction.

10 and 15 Mark

1. Explain hydroelectric power plant with schematic diagram.
2. Explain power factor correction using capacitors.
3. Explain sag calculation and its significance.
4. Explain distribution schemes, insulators, cables and substations.

Model Question Paper

Time: 3 Hours · **Maximum Marks:** 100

1. Define base load and peak load.
2. Explain hydroelectric generation with diagram.
3. Calculate hydel power for $Q=25\ m^3/s$, $H=60$ m and $\eta=0.82$.
4. Calculate capacitor kVAr for a 250 kW load at 0.75 improved to 0.95.
5. Explain overhead line components and sag.
6. Explain underground cable construction and substation single-line diagram.

Answer Key

Hydel equation

$P=9.81QH\eta$ kW.

Hydel problem

$P=9.81\times25\times60\times0.82=12066.3$ kW.

12.07 MW

Capacitor problem

$Q_c = 250(0.882 - 0.329)$.

138.25 kVAr